CSWDO MIS: Queue Management Module — Master Group Brief & Technical Guide

This complete functional, database, and user interface blueprint covers the **Queue Management Module** for the **City Social Welfare and Development Office (CSWDO) Management Information System**. Every instruction below is designed to integrate our new **Online Queue ID (**OL-"2026-current date"**)** and **Walk-In Queue ID (**WI-"2026-current date"**)** formats while keeping full compatibility with the rest of the class.

1. Golden Rules & Module Boundaries

To ensure our module never breaks the workflows of other groups, every member must follow these core rules:

- **Create and Align the Database First:** The entire class must create and use a unified database first to avoid conflicts across modules.

- **Never Break the Control Number Requirement:** The project specifies that the final output of the client application is a **Control Number**, which the client shows to the employee along with hard copies of their requirements. The employee inputs this Control Number to view and validate client information, the queue operates based on the Control Number, and the same Control Number is used later for customer feedback.

- **Seamless Pairing of Control Number + Queue ID (**OL **&** WI**):**

    - To make the live queue display clear without breaking the **Control Number** lookup, every queued client will have their **Control Number** linked 1:1 with a standardized **Queue ID**:

        - **Online Queue ID:** OL-"2026-current date" (e.g., OL-2026-1001-01)

        - **Walk-In Queue ID:** WI-"2026-current date" (e.g., WI-2026-1001-01)

    - *Note for Class Integration:* If the class agrees to use OL-2026-[current date] and WI-2026-[current date] as the actual format of the control_number itself, our system supports it automatically. If the Registration Group uses a different control_number format, our queue_logs table stores **both** control_number and queue_code (OL-… / WI-…) so neither module breaks.

- **Enforce Daily Slot Capacities:** Online scheduling is strictly divided into **AM (30 slots)** and **PM (30 slots)**. Our queue system must handle situations where all scheduled applicants arrive at the same time, ordering them based on their Control Number.

- **Support Both Online and Walk-In Clients + User Counter:**

    - **Online Clients (**OL-2026-current date**):** Arrive with their Control Number and physical requirement documents for employee validation.

    - **Walk-In Clients (**WI-2026-current date**):** Follow the same verification and validation process, after which the system shows a status of **"for interview"**.

    - **Total Number of Users:** The system must track and display the number of users (both walk-in and online).

- **Strict UI & Web Development Standards:**

- The design must use a **Blue and White** color palette and include the official **Pasig Logo**.

- We must apply proper principles of web development throughout our module.

- Any formal header or display referencing the office address should use **Caruncho Avenue, Barangay San Nicolas, Pasig City 1600 Metro Manila**.

- **What We Do NOT Build (Other Groups' Scope):**

  - We do **not** build the Client Account Creation (with Pasig Pass injection) or the Assistance Application Form (Medical, Financial, Burial, Balik Probinsiya, etc., excluding educational assistance, with the 3-month application limit). We only **read** the resulting Control Number, client info, schedule slot, and attachments.

  - We do **not** build the pre-defined interview discussion questions, the Social Case Study Report form, the Intake Sheet form, or the Customer Satisfaction Survey form. We only **trigger** the status changes, Intake Sheet timestamps, and Admin-to-Staff notifications that connect them.

2. Format & Logic for OL **and** WI **Queue IDs**

Our module will automatically generate and format the Queue ID upon check-in/validation using the current date in 2026:

Client Type	Prefix	Format Pattern	Example Output (Oct 1, 2026)	Slot / Capacity Rule
Online Client	OL-	OL-2026-MMDD-[Seq] *(or* OL-2026-MM-DD-[Seq]*)*	OL-2026-1001-001	Capped at **30 AM slots** and **30 PM slots** (001 to 030 per session)
Walk-In Client	WI-	WI-2026-MMDD-[Seq] *(or* WI-2026-MM-DD-[Seq]*)*	WI-2026-1001-001	Increments dynamically and counts total walk-in users for the day

- **How Simultaneous Arrivals Are Ordered:** When multiple scheduled applicants show up at the office at the same time, the system queues them based on their **Control Number** and assigns their corresponding OL-2026-[current date] sequence.

- **Lookup Flexibility:** On the Employee and Admin screens, searching **either** the control_number **or** the OL-2026-… / WI-2026-… ID will pull up the exact same client record.

3. Step-by-Step System Flow (For Our Flowchart Deliverable)

The project requires a **flowchart for the system flow**. Use this exact 6-step sequence to draw our flowchart and program the backend logic:

1. **Client Arrival & Control Number Input:**

- **Online Client:** Shows their **Control Number** and hard-copy requirements to the employee. The employee inputs the Control Number into the system.

- **Walk-In Client:** Undergoes the same verification and validation process on-site, where the system tracks the number of users.

2. **Document Validation & Queue ID Generation (**OL **/** WI**):**

    - Studying the screen, the employee views the client's personal information and uploaded attachments and validates them against the physical hard copies.

    - The employee sets the status to **"for Interview"** (or **"Approved"** if no interview is required).

    - When set to **"for Interview"**, the system generates the client's live Queue ID—OL-"2026-current date" for online scheduled applicants (within the 30 AM or 30 PM slots) or WI-"2026-current date" for walk-in applicants—ordered by their Control Number.

3. **Live Queueing & Calling the Client to Interview:**

    - The client waits while watching the Live Public Queue Monitor displaying OL-2026-... and WI-2026-... alongside their Control Number.

    - When the interviewer (Social Welfare Officer) clicks **"Call Next"** and starts the interview, the system automatically logs the **Date**, **Interviewer**, and **Time Interview** fields required on the CSWDO Intake Sheet.

4. **Post-Interview & Admin Evaluation Trigger:**

    - Once the interviewer finishes the pre-defined interview questions / Intake Sheet / Social Case Study Report assessment, the queue status updates for the **Admin Side**, which displays the status of all clients.

    - Our module triggers an **alert, notification, and message** on the Admin Side indicating that the record needs to be scanned and evaluated so the Admin can click **Approve**.

5. **Admin Approval & Real-Time Staff Notification:**

    - The Admin reviews the evaluation and clicks **Approve**.

    - Immediately after the Admin approves, the system sends a **notification back to the staff** and updates the status to **Approved**.

6. **Report Generation, Release Timestamp & Survey Hand-off:**

    - The notified staff generates the final report that only needs to be signed.

    - When the signed report is handed to the client, the staff clicks **"Release Client"**, which logs the **Time Released** for the Intake Sheet, closes the active OL- / WI- queue ticket, and passes the **Control Number** to the injected **Customer Satisfaction Survey** for customer feedback.

4. Shared Database Design (SQL Blueprint)

Because the class must create a unified database first to avoid conflicts, share this schema with the class database lead.

A. Columns Required in the Shared applications **Table**

Ensure the main applications table created by the class includes:

- control_number (VARCHAR(50), Primary/Unique Key) — Generated after application submission and used for queue lookup and customer feedback.
- client_type (ENUM('Online', 'Walk-In')) — Identifies whether the client applied online or walked in.
- assistance_type (VARCHAR(50)) — Medical, Financial, Burial, Balik Probinsiya, etc. (excluding educational assistance).
- schedule_date (DATE) & schedule_slot (ENUM('AM', 'PM', 'Walk-In')) — Enforces the 30 AM slots and 30 PM slots.
- application_status (ENUM('Pending Validation', 'for Interview', 'For Admin Approval', 'Approved', 'Released')) — Displays client status for Employee and Admin views.

B. Our Queue Management & Notification Tables

SQL

-- 1. Main Queue Management Table (Stores both Control Number and OL/WI Queue IDs)

```
CREATE TABLE queue_logs (
  queue_id INT AUTO_INCREMENT PRIMARY KEY,
  queue_code VARCHAR(30) NOT NULL UNIQUE,    -- Format: OL-2026-MMDD-001 or WI-2026-MMDD-001
  control_number VARCHAR(50) NOT NULL,       -- Master Control Number from client application
  client_type ENUM('Online', 'Walk-In') NOT NULL,
  schedule_slot ENUM('AM', 'PM', 'Walk-In') NOT NULL,
  queue_number INT NOT NULL,                 -- Daily sequence number (1-30 for AM/PM, 1-N for Walk-In)
  queue_status ENUM(
    'Waiting',
    'Called',
    'In Interview',
    'For Admin Evaluation',
    'Approved - Ready for Release',
    'Released',
    'No Show'
  ) DEFAULT 'Waiting',
  assigned_window VARCHAR(50) DEFAULT NULL,
  interviewer_name VARCHAR(100) DEFAULT NULL, -- Maps to INTERVIEWER on CSWDO Intake Sheet
  queue_date DATE NOT NULL,                  -- Maps to Date on CSWDO Intake Sheet (e.g., 2026-10-01)
  time_checked_in DATETIME DEFAULT CURRENT_TIMESTAMP,
  time_interview TIME DEFAULT NULL,          -- Maps to Time Interview on CSWDO Intake Sheet
```

```
  time_released TIME DEFAULT NULL,          -- Maps to Time Released on CSWDO Intake Sheet
  FOREIGN KEY (control_number) REFERENCES applications(control_number) ON UPDATE CASCADE
);

-- 2. Admin & Staff Alert, Notification, and Message Table
CREATE TABLE system_notifications (
  notification_id INT AUTO_INCREMENT PRIMARY KEY,
  control_number VARCHAR(50) NOT NULL,
  queue_code VARCHAR(30) NOT NULL,          -- Displays OL-2026-... or WI-2026-... in alerts
  sender_role ENUM('Employee', 'Interviewer', 'Admin') NOT NULL,
  recipient_role ENUM('Employee', 'Interviewer', 'Admin') NOT NULL,
  alert_type ENUM('Alert', 'Notification', 'Message') NOT NULL DEFAULT 'Notification',
  alert_title VARCHAR(120) NOT NULL,
  message_body TEXT NOT NULL,
  is_read BOOLEAN DEFAULT FALSE,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (control_number) REFERENCES applications(control_number) ON UPDATE CASCADE
);
```

C. Helper Logic for Generating OL-"2026-current date" **and** WI-"2026-current date"

Give this logic to our backend programmer so the ID format is always consistent:

- **Prefix Determination:** If client_type == 'Online', prefix is OL. If client_type == 'Walk-In', prefix is WI.

- **Date Stamp:** Format the current date starting with 2026 (e.g., 2026-10-01 or 2026-1001).

- **Sequence Appending:** Count today's existing records for that client_type (and schedule_slot if Online, enforcing <= 30 for AM and <= 30 for PM), add 1, and pad to 3 digits:

    - **Online Example:** OL-2026-10-01-001 (or OL-2026-1001-001)

    - **Walk-In Example:** WI-2026-10-01-001 (or WI-2026-1001-001)

5. Screens & Features Our Group Must Build

All screens must follow the **Blue and White** color scheme, display the **Pasig Logo**, and reflect good web development principles.

Screen 1: Employee Validation & Check-In Desk

- **Control Number Lookup Bar:** The employee inputs the **Control Number** presented by the client.

- **Information & Attachment Viewer:** Shows the client's personal info, assistance type, and uploaded attachments so the employee can validate them against the client's physical hard copies.

- **Walk-In Registration Trigger:** Allows staff to verify and validate a walk-in client using the same process.

- **Validation Action Buttons:**

    - **"Validate & Set to 'For Interview'"** — Updates status to "for interview", assigns the OL-"2026-current date" or WI-"2026-current date" Queue ID ordered by Control Number, and places the client into the active queue.

    - **"Validate & Approve Directly"** — Sets status to "Approved" if no further interview is required.

Screen 2: Live Public Queue Monitor (Waiting Area TV View)

- **Header:** Blue and White banner with the **Pasig Logo**, office address (**Caruncho Avenue, Barangay San Nicolas, Pasig City 1600 Metro Manila**), and live clock.

- **Now Serving Display:** Large cards showing:

    - **Queue ID:** OL-2026-10-01-001 or WI-2026-10-01-001

    - **Control Number:** Displayed underneath so the client can match either code.

    - **Assigned Window / Interviewer**.

- **Waiting Queue Columns:**

    - **Online Queue (**OL-2026-current date**):** Ordered by Control Number for simultaneous arrivals.

    - **Walk-In Queue (**WI-2026-current date**):** Active walk-in applicants waiting for interview.

- **Live Capacity & User Counter Bar:**

    - AM Online Slots: X / 30

    - PM Online Slots: X / 30

    - Total Walk-In Users Today: X

    - Total Active Users in Queue: X

Screen 3: Interviewer (Social Welfare Officer) Queue Controller

- **Call Next Client Button:** Calls the next OL-2026-… or WI-2026-… ticket in line based on Control Number.

- **Start Interview Button:** Automatically logs the **Date**, **Interviewer**, and **Time Interview** for the Intake Sheet.

- **Submit for Admin Approval Button:** Updates the client status on the Admin side and dispatches an alert, notification, and message to the Admin.

- **Generate Signable Report & Release Client Button:** Activated once the Admin approves and staff receives the notification. Generates the report that only needs to be signed, logs **Time Released** for the Intake Sheet, and releases the **Control Number** for the Customer Satisfaction Survey.

Screen 4: Admin & Staff Alert / Notification / Message Widget

- **Admin Side Widget:** Displays the status of clients alongside an **alert, notification, and message** prompting the Admin to scan/evaluate the record and just click **Approve**.

- **Staff Side Notification Toast:** Automatically notifies the staff the moment the Admin clicks **Approve** (e.g., *"Notification: Queue ID OL-2026-10-01-001 (Control No. X) has been APPROVED by Admin. Ready to generate report for signature."*).

6. Non-Coding Deliverables Required by the Project Rubric

In addition to our code, the instructions require two specific deliverables that our group must prepare for our section:

1. **System Flowchart:** A visual flowchart illustrating the 6-step Queue Management workflow from Control Number input and OL/WI queue generation to Admin notification and report release.

2. **Deployment Expenses Report Table:** The project requires showing an expenses report—simply displayed as a table—estimating how much will be spent for deployment. Include this table in our presentation/UI:

Expense Item	Role in Queue Management Deployment	Estimated Cost (PHP)
Shared Cloud Web & Database Server	Hosts the CSWDO MIS database, queue controller, and real-time notification service	₱1,500.00 / month
43" Smart LED TV Monitor	Public Waiting Area display showing OL-2026-… and WI-2026-… live queue	₱14,500.00 (One-time)
Thermal Queue Ticket Printer	Prints WI-"2026-current date" and Control Number slips for Walk-In clients	₱2,800.00 (One-time)
Wall Mount Bracket & HDMI Cabling	Physical installation of the queue monitor at the CSWDO waiting lobby	₱850.00 (One-time)

7. Group Member Task Assignment

- **Member 1 (Database & Backend Queue Logic):** Coordinates with the class to finalize the shared database first; writes the generator for OL-"2026-current date" and WI-"2026-current date"; enforces the **AM (30 slots)** and **PM (30 slots)** limits; and logs time_interview and time_released for the Intake Sheet.

- **Member 2 (Employee Validation & Interviewer Controller UI):** Builds the Blue-and-White Employee Control Number Lookup & Attachment Validation screen, the Walk-In user counter, and the Interviewer Queue Control panel.

- **Member 3 (Public Queue Display & Real-Time Notifications):** Builds the Live Public Queue Monitor (OL- and WI- display) and programs the Admin-to-Staff **alert, notification, and message** system when clicking **Approve**.

- **Member 4 (Flowchart, Expenses Table & Integration QA):** Draws the System Flowchart, codes the HTML Deployment Expenses Report Table, and verifies that

the **Control Number** passes cleanly from Client Registration → Queue Management → Report Signing → Customer Satisfaction Survey.